SYSADMIN & SECURITY HANDBOOK | SYSCOHADA REVISED (AUDCIF) | IFRS STANDARDS



Figure 1: LedgerCore Health Dual-Standard Architecture — Unified Double-Entry Core connecting SYSCOHADA Revised (OHADA), IFRS Presentation, Biomedical Asset Lifecycle, and Hospital Cash Desks.

1. System Deployment & Environment Setup

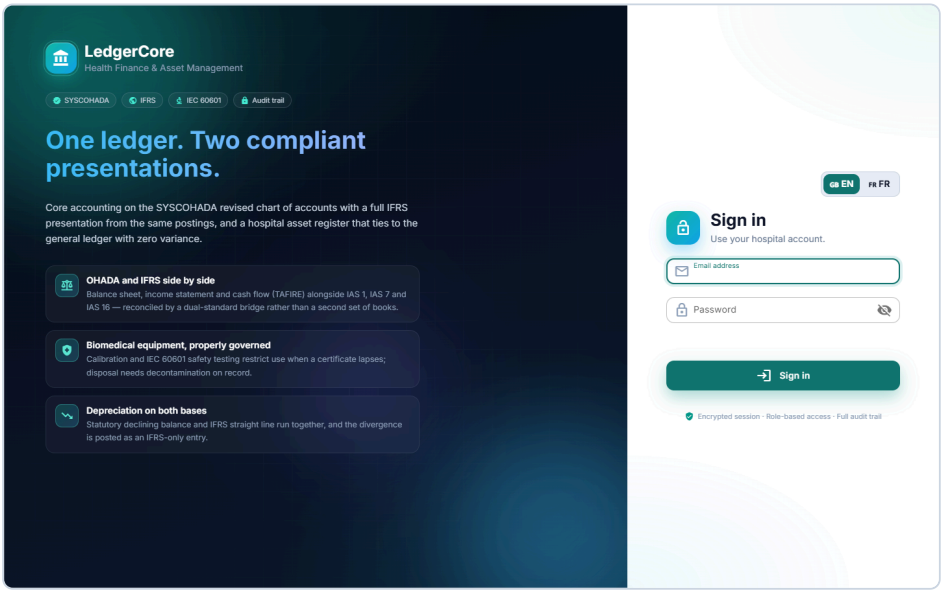


Figure 2: Authentication Portal — Language Selection, Encrypted Session Handshake, and RBAC Profiles.

LedgerCore Health operates as a high-performance decoupled architecture: - ⚡ **Backend API:** Node.js (Express, ES Modules, MySQL2/MariaDB connection pool) on port `4700`. - 🌐 **Frontend UI:** Python 3.10+ (NiceGUI, Quasar/Vue3, TailwindCSS) on port `8700`. - 🗄️ **Database:** MariaDB 10.5+ / MySQL 8.0 on port `3306` (Database: `ledgercore`).

Step-by-Step Procedure: Starting Services on Windows Server

1. 🗄️ **Database Service:** Ensure MariaDB / MySQL is running via Windows Services or XAMPP Control Panel.
2. ⚡ **Launch Backend API:**

```
cd "C:\Core Accounting and asset\backend"
npm start
```

Health Check Verification: Open `http://127.0.0.1:4700/api/health` — must return status `healthy`.

3. 🌐 **Launch Frontend Service:**

```
cd "C:\Core Accounting and asset\frontend"
python main.py
```

UI Access: Open `http://127.0.0.1:8700` in Google Chrome or Microsoft Edge.

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2. Step-by-Step Procedure: User Provisioning & Granular RBAC Permissions

Under **Administration → Users & Permissions** (`/users`): 1. 👤 Click **+ Add User**. 2. 📝 Enter email, full name, phone number, and default language (`en` or `fr`). 3. 🔑 Set the initial temporary password and enable **Force password change on first sign-in**. 4. 🛡️ Assign one or multiple role profiles: - `admin` : System configuration, organization settings, user management. - `cfo` : Full financial posting, period close, depreciation, and reporting. - `accountant` : Voucher preparation, open-item matching, bank reconciliation. - `asset_manager` : Asset register, maintenance scheduling, physical inventory. - `biomed_tech` : Work order execution, IEC 60601 testing, calibration logging. - `services_tech` : Errands, shopping lists, vehicle fleet logs. - `ward_head` : Departmental asset view and breakdown ticket submission. 5. 💾 Click **Save User**.

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3. Step-by-Step Procedure: Organization Settings & Statutory Posting Rules

Under **Administration → Organization Settings** (`/settings`): - 💱 **Functional Currency:** Default `XAF` (Central African CFA franc) or `XOF` . - 📒 **Accounting Frameworks:** Select `OHADA` , `IFRS` , or `Both` (Dual-Standard Mode). - 🔢 **Posting Length (Account Code Digits):** Set statutory account digits (4, 6, or 8 digits). Note: Once transactions exist, posting length is permanently locked to preserve historical audit integrity. - ⏳ **Suspense / Clearing Account:** Default `471100` for unclassified banking receipts awaiting allocation.

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4. Cryptographic Seal Verification & Forensic Audit Trail



Figure 3: 12-Step Month-End Closing Procedure & Cryptographic SHA-256 Hash Chaining in LedgerCore Health.

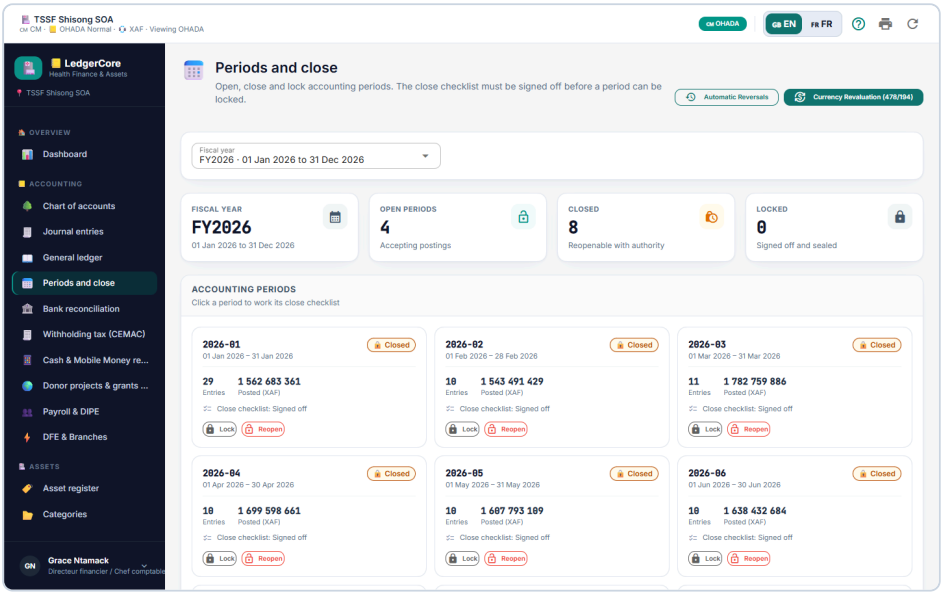


Figure 4: Period Management Screen — Verifying Audit Seal Integrity and Permanent Chained Hashes.

LedgerCore incorporates an immutable **SHA-256 blockchain-style seal** on journal entries: - 🔗 Every posted entry hashes the previous entry's hash, timestamp, user ID, account lines, and monetary amounts. - 🛡️ Under **Reporting → Audit Seal Verification**, run the integrity scanner: - If any row in the `journal_entries` or `journal_lines` database table was altered outside the application (e.g., direct SQL injection or manual database update), the cryptographic verification fails immediately with the exact tampered row number highlighted in red.

5. Automated Database Backup & Disaster Recovery

Automated Daily Dump (PowerShell Script):

```
$Date = Get-Date -Format "yyyy-MM-dd_HH-mm"
$BackupFile = "C:\Backups\ledgercore_backup_$Date.sql"
mysqldump.exe -u root -p"your_password" --single-transaction --quick --routines --triggers ledg
```

Restoration Protocol:

```
mysql.exe -u root -p"your_password" ledgercore < "C:\Backups\ledgercore_backup_2026-10-18.sql"
```